

Robust Super Twisting Based Sliding Mode Control for a 2-DOF Nonlinear Helicopter Model

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Abstract—This paper presents a robust control strategy, namely Sliding Mode Control (SMC), to control a two-degree-of-freedom (2DOF) helicopter model. This model is an inherently unstable, nonlinear, and multivariable system with coupling effects between two axes. Due to the mentioned characteristics, helicopter control remains a challenging problem, especially in the presence of external disturbances, parameter perturbation and model uncertainties. In order to overcome some of these difficulties, we developed a sliding mode controller based on the super-twisting algorithm (STSMC). It represents a modification of the standard SMC with the aim of improving its control capabilities. The effectiveness and robustness of the proposed STSMC were verified through simulations and experiments. The results obtained were compared with the same achieved by a previously developed PID controller. The STSMC significantly outperforms the PID controller used in terms of stability of the system, tracking error, and overall control performance in the presence of system uncertainties and coupling effects between two axes, the azimuth and elevation.

I. INTRODUCTION

Nonlinear systems play a vital role in modern control engineering, since all plants in practice are mainly nonlinear in nature [1]. The study of nonlinear systems is attracting increasing attention from researchers to explore the use of nonlinear control methods as a way of coping various issues encountered when using linear control of nonlinear processes. [2], [3]. One such nonlinear, inherently unstable, dynamic system is a helicopter model with two degrees of freedom (2DOF), which is the subject of this paper. In the last two decades, helicopters have received increased attention because they offer a wide range of various applications, including the transport of people and cargo, search and rescue missions, military and police operations, medical transportation, firefighting operations, aerial observations, and air pollution monitoring, among others. These applications are possible because of the helicopter's ability to take off and land vertically, to hover, and to fly forward, backward, and laterally.

Regarding a flight control of the helicopter system, conventional strategies involve linearizing the dynamics of the nonlinear helicopter model around a set of preselected equilibrium points. These control techniques are mainly based on SISO (single-input-single-output) approaches with PID control laws. However, this requires multi-loop controllers, which make their design inflexible and difficult to tune controller parameters. Afterwards, in our paper we consider a multi-input-multiple-output (MIMO) nonlinear helicopter

model which is used as a plant in modern control theory [4]. It includes the use of advanced controllers, such as gain scheduling [5], linear quadratic regulator (LQR) control [6], loopshaping-based controller [7], model predictive control (MPC) [8], robust sliding mode control [9], nonlinear h_∞ control [10], various types of adaptive-fuzzy [11], [12] control, backstepping [13], neural network [14], etc.

The controlled system used in our paper is HUMUSOFT's CE 150 helicopter model. It is a good case study of nonlinear system control, suitable for testing any control techniques aimed at supervision of nonlinear behavior. This MIMO model consists of three potential inputs (voltages), one input for each of the two DC motors responsible for driving the propellers, and one for manipulating the center of gravity. Motor input voltages have a linear relation with the torques generated. The output of the helicopter model represents the azimuth and elevation angles, which are measured by incremental encoders. The helicopter model shows strong cross-coupling effects between the elevation and azimuth dynamics. This means that the operation of one motor affects the performance of another. Among other factors, cross-coupling increases the difficulty of proper control design. The system is fixed on a base and contained in a cage in order to minimize the risk of damage to both the working environment and the system. Direct communication with the physical model of the CE 150 helicopter is established using Real-Time Toolbox, which is an integral part of Matlab/Simulink environment. Information about the system was extracted from [15], which is referred to for more information.

In this paper, the STSMC is developed with the aim of simultaneously tracking the azimuth and elevation position references of the nonlinear multivariable helicopter model. This controller deals with system uncertainties caused by cross-coupling action between the azimuth and elevation axes. In addition, the proposed STSMC can cope effectively with the nonlinear characteristics of the used 2DOF helicopter model. In order to overcome the above difficulty and improve the helicopter tracking performance, some modifications have been made to the SMC.

The paper is structured as follows: Section II describes and explains the used 2DOF multivariate and nonlinear helicopter model. In Section III the second-order sliding mode controller (STSMC) is developed with the aim of coping with high nonlinearities and cross-coupling effects of the helicopter. The evaluation of the control performance of the proposed STSMC compared to the previous designed PID controllers is done through both simulations and experiments

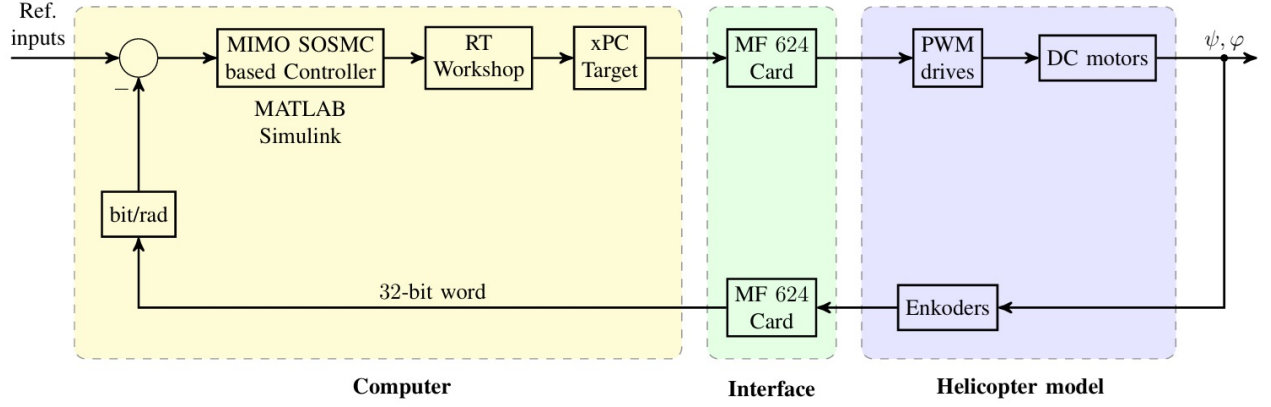


Fig. 1. Proposed real-time control scheme for helicopter model.

in Sections IV and V. Section VI summarizes our findings and gives a brief exposition of future research directions.

II. CONTROL SYSTEM DESCRIPTION

The proposed real-time control system for the Humusoft CE150 2DOF helicopter model is shown in Figure 1. The overall system includes: i) PC based STSMC controllers, ii) interface module, and iii) real-helicopter system. More details on the control interface used can be found in [14]. The helicopter is modeled for purposes of controller synthesis and performance tracking validation. The helicopter is a rigid body with two DC motors driving the main and side rotors, power amplifiers, and encoders as sensors (Figure 2).

The helicopter model used has two degrees of freedom: elevation angle ψ representing rotation around the horizontal axis and azimuth angle ϕ representing rotation around the vertical axis. The axes of the main and side propellers are orthogonal. The helicopter model is a MIMO system with two input signals (voltage of the main motor u_1 and voltage of the side motor u_2 with operation ranges $[0, 0.6]$ and $[-0.3, 0.3]$, respectively) and two output signals (elevation and azimuth with operation ranges $[-45, 45]$ degree and $[-130, 130]$ degree, respectively).

The equations of helicopter motion in the vertical and horizontal planes are derived using the Euler-Lagrange equations

given by eElevation dynamics:

$$a_\psi \ddot{\psi} = \tau_1 + \tau_\phi - \tau_{f1} - \tau_m + \tau_G, \quad (1)$$

$$\tau_m = mgl \sin \psi, \quad (2)$$

$$\tau_\phi = ml\dot{\phi}^2 \sin \psi \cos \psi, \quad (3)$$

$$\tau_{f1} = C_\psi \text{sign } \dot{\psi} + B_\psi \dot{\psi}, \quad (4)$$

$$\tau_G = k_G \dot{\phi} \omega_1 \cos \psi, \text{ for } \dot{\phi} \ll \omega_1, \quad (5)$$

and azimuth dynamics:

$$a_\phi \ddot{\phi} = \tau_2 - \tau_{f2} - \tau_r, \quad (6)$$

$$\tau_{f2} = C_\phi \text{sign } \dot{\phi} + B_\phi \dot{\phi}, \quad (7)$$

$$a_\phi = a_\psi \sin \psi. \quad (8)$$

where: a_ψ is moment of inertia around horizontal axis, τ_1 is moment produced by the main motor propeller, τ_ϕ is centrifugal torque, τ_{f1} is Coulomb and viscous friction torques, τ_m is gravitation torque, τ_G is gyroscopic torque, m is mass of the helicopter body, g is gravitational acceleration, l is distance from z axis to main motor axis, ω_1 is angular velocity of the main propeller, k_G is gyroscopic coefficient, B_ψ is viscous friction coefficient, C_ψ is Coulomb friction coefficient, a_ϕ is moment of inertia around vertical axis, τ_2 is moment produced by the side motor propeller, τ_{f2} is Coulomb and viscous friction torques, τ_r is reaction torque of the main motor, ω_2 is angular velocity of the side propeller, B_ϕ is viscous friction coefficient, C_ϕ Coulomb friction coefficient.

In addition, the dynamics of the main and tail motor model can be described as a second-order transfer function[16]:

$$\frac{\omega_i(s)}{u_i(s)} = \frac{1}{(T_i s + 1)^2}, \quad (9)$$

where $\omega_i(s)$ and $u_i(s)$ are Laplace transforms of $\omega_i(t)$ and $u_i(t)$ respectively, while T_i is the time of the main motor ($i = 1$) and the tail motor ($i = 2$).

Detailed description of the helicopter used and the identification of its individual parameters can be found in [17]. It was necessary to identify 14 unknown parameters. The

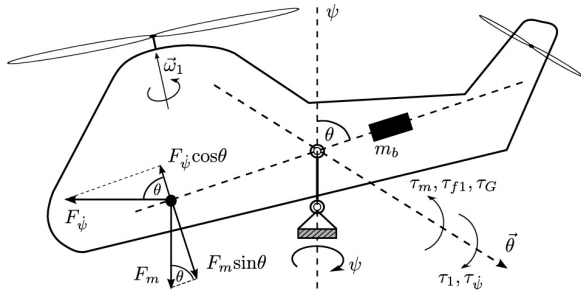


Fig. 2. Simplified helicopter model with two inputs and two outputs.

identification procedure was performed experimentally, but there is some uncertainty about the accuracy of all parameters. Also, it is necessary to take into account an existing strong cross-coupling and significant nonlinearities in the model itself. Since the mathematical model of the system has some uncertain parameters, it is needed to use some variant of the robust control. Our paper focuses on one of the robust control strategies, more precisely the sliding mode control.

III. SLIDING MODE CONTROL

One of the promising techniques for designing robust control laws is the sliding-mode control. It represents a nonlinear control approach which provides robustness to parameter variations, reduced order of sliding-mode dynamics, and finite-time convergence to zero for sliding-mode variables [18]. The main idea of SMC is the design of a surface in the state space of the system that satisfies the required closed-loop dynamics, and the synthesis of a control law that steers the trajectory of the system towards this surface. Within the proposed SMC control law, the sliding variable is associated with the tracking error and its derivatives as follows: [19]:

$$\sigma = \sigma(e, \dot{e}, \dots, e^k) \quad (10)$$

The function $\sigma = 0$ defines a sliding surface (or manifold). The objective is to determine a control law that forces the system's trajectory to a region in which the sliding variable is equal to zero [20]. This region is called the sliding surface (or manifold). It is crucial to choose the SMC control law so that the sliding surface exists and is reachable along system trajectories in finite time, so-called the reaching time.

After designing a sliding surface, the next step in SMC design is to derive a control law that guaranties that σ converges to zero in finite time. If a well-defined controller is able to drive the sliding variable σ to a neighborhood of zero, the sliding variable will consistently rise and fall around zero, resulting in a phenomenon called chattering. It causes an undesirable behavior of high-frequency control switching, resulting in a high-frequency vibration of the system [21]. Chattering can induce stress on the mechanical and electrical components of the structure, which can eventually destabilize the entire system [22]. In order to reduce the effects of chattering, continuous functions can be used to mimic the *sgn* function and replace its discontinuity.

In order to obtain a compromise between robustness to uncertainties and avoidance of chattering, we developed the second-order SMC (SOSMC). It employs the "Super-Twisting" algorithm, with additional differentiators in the form of a multivariable controller. This algorithm is taken from [19] and is expressed by the following equations:

$$u = \lambda \sqrt{|\sigma|} \text{sgn}(\sigma) + w \quad (11)$$

$$\dot{w} = W \cdot \text{sgn}(\sigma) \quad (12)$$

$$\lambda = \sqrt{U} \quad W = 1.1U \quad (13)$$

where U is a positive constant which is empirically determined, usually via simulation testing if the system's model is available. The parameter is calibrated using the trial and error

process until the desired results are achieved. The control signal u manages the outputs of the system based on the output tracking error in the following manner:

$$\sigma = \sigma(e, \dot{e}) = e + \dot{e} \quad (14)$$

$$e(t) = r(t) - y(t) \quad (15)$$

where: $r(t)$ - reference input signal, $y(t)$ - output signal, and $e(t)$ - tracking error.

The major benefit of the formulated control approach is the simplicity of its implementation, since only one parameter needs to be adjusted during the refinement phase. In order to illustrate the effectiveness and feasibility of the given control law, both simulation and experimental results are addressed.

IV. SIMULATION RESULTS

The nonlinear MIMO helicopter plant model as well as the proposed multivariable STSMC are developed in the Matlab/Simulink environment. In order to determine the appropriate parameters of the STSMC controller, the simulations were repeated for their different values, and thus the obtained system responses were compared with the desired system output. The same procedure was carried out for the PID controller[5], whose control performance will be compared to the STSMC controller developed.

Figures 3 and 4 show the results obtained by the PID and STSMC controllers for azimuth and elevation angles. In these graphs, the black line represents the reference signal, the blue line shows the output of the PID controlled system, and the output of the STSMC controlled system is shown by the red line. The control performance of employed controllers was evaluated with a complex reference signal consisting of a series of consecutive step excretions of different amplitude values. This form of input signal is chosen to simulate sudden changes that the helicopter might encounter in its operating environment.

It is obvious from the figures shown that the STSMC controller has no overshoot/undershoot and also reduces considerably the settling time and steady-state errors in comparison to the PID controller. A more significant deviation in settling time is observed with the PID controller, especially with azimuth control. The responses of both azimuth and elevation angles with the PID controller indicate their significant deviation from the reference signal within the duration of the first two-step signal. There are some minor overshoots and undershoots in the elevation angle response of the PID controller (Figure 3), while the azimuth angle response with this controller is more oscillatory than with the STSMC (Figure 4). It is important to note that grater over/undershoots in the system responses can have catastrophic consequences for systems that need flight maneuvers such as take-off and landing [23]. Moreover, a system output that uses the STSMC controller satisfactorily tracked the reference signals without any significant delay.

In conclusion, the designed STSMC controller significantly outperforms the baseline PID controller in numerous quantities, most notably in its response time, rising time,

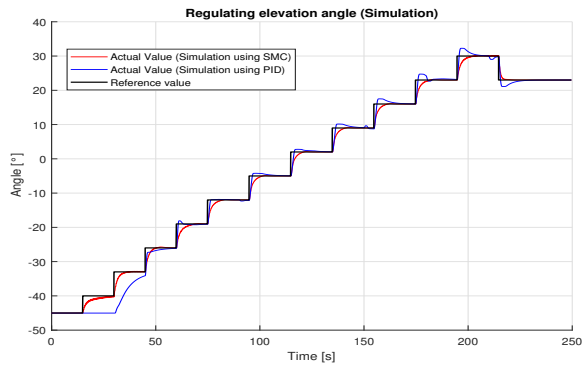


Fig. 3. Simulation results (elevation angle).

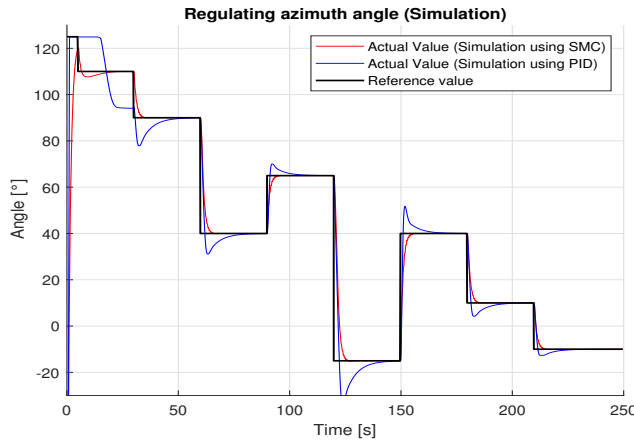


Fig. 4. Simulation results (azimuth angle).

and settlement time. In addition, the STSMC system's response has far less fluctuations than its PID counterpart, and remains steady throughout the observed time interval. The previous conclusions provide strong evidence to suggest that the STSMC controller will perform correctly on the actual physical system.

V. EXPERIMENTAL RESULTS

This section illustrates the experimental implementation of the STSMC control strategy on the Humusoft CE 150 physical helicopter system. In addition, the PID-controller is used as a comparison to the proposed STSMC. In this experimental evaluation of the controllers used, the guidelines outlined in [24] were followed. The experiment involves integrating the developed controllers into a real system, comparing sensor outputs with reference values, and performing multiple execution sequences to ensure repeatability of the experiment. The system was tested under laboratory conditions, minimizing the influence of external factors.

In Figures 5 and 6 the time response of the STSMC-based control strategy is compared to the PID controller. Figure 5 shows the pitch angle and Figure 6 shows the yaw angle. Both plots illustrate a significant improvement of the STSMC controller compared to the PID-controller. The azimuth and elevation angle time responses of the PID controller are more

oscillatory than those of the STSMC controller, especially in the time interval between 50 and 110 seconds. The STSMC significantly reduces oscillations in azimuth and elevation angle response, as well as a steady-state error trading overshoot for a faster response. Moreover, although there is an initial delay in following the reference values for both the elevation and azimuth angles, the STSMC outperforms the PID controller in this regard. A particular advantage of the STSMC is that it almost eliminated the coupling between the azimuth and elevation angles. Furthermore, the STSMC successfully copes with nonlinearities of the helicopter model and compensates the influence of the unmodeled dynamics of the helicopter model, which is difficult to achieve with the used PID-controller, especially with a physical model of helicopter.

VI. CONCLUSIONS

The sliding-mode-based control strategy for a nonlinear two-DOF helicopter model as an inherently unstable system was presented. The designed STSMC controller guarantees the stability of the system during steady state and the desired performance despite modeling uncertainties as well as coupling between angles, where the system model is not accurately described. Numerical simulations and experiments are provided to illustrate the capability of the STSMC control law. The results obtained heavily imply that the STSMC approach significantly reduces the tracking error and oscillations in both azimuth and elevation angles, and overall provides better control system performance than previously designed PID-controller. The superiority of STSMC over PID-controller is especially evident in the real-time implementation, ie. in physical experiments. Potential future work could focus on testing other robust control methods or different types of SMC control strategies. An interesting topic will be the integration of adaptive control approaches with robust control strategies.

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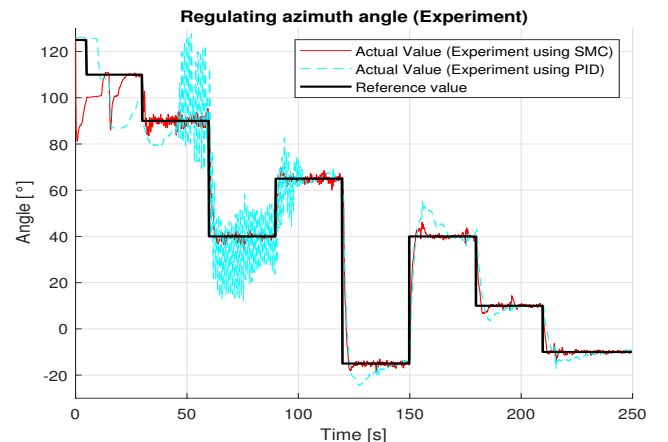


Fig. 5. Experimental results (azimuth angle).

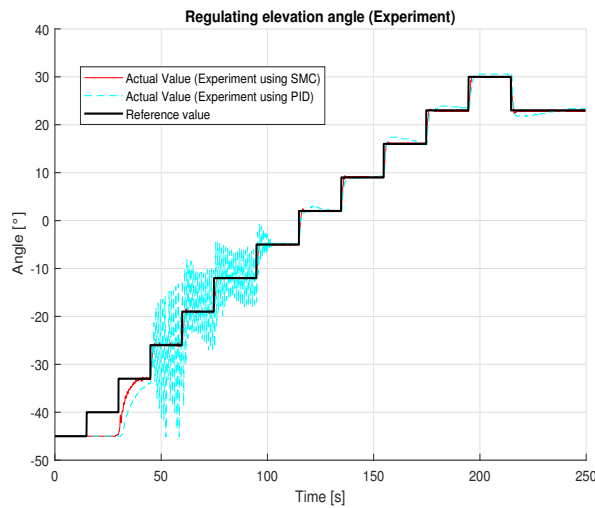


Fig. 6. Experimental results (elevation angle).

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